

FITBUDDY – AI FITNESS PLAN GENERATOR USING GEMINI MODELS

1. Brainstorming & Ideation Phase

The Brainstorming and Ideation Phase is the initial stage of the FitBuddy project. The main purpose of this phase is to identify a useful real-world problem and develop an idea for solving it using Artificial Intelligence. Fitness planning is an area where people often need personalized workout routines, nutrition suggestions, and recovery guidance. A general workout plan may not be suitable for every individual because people's age, weight, fitness goals, and workout intensity preferences are different.

During the brainstorming process, the idea of developing FitBuddy – AI Fitness Plan Generator using Gemini Models was proposed. The objective of FitBuddy is to create personalized fitness plans based on the information provided by the user. Instead of manually preparing a workout plan, the application uses Google's Gemini AI models to generate a structured workout plan.

The main features identified during the ideation stage were user information collection, AI-generated workout plans, nutrition and recovery recommendations, user feedback, and automatic workout-plan updates. The system should allow users to enter details such as User ID, Name, Age, Weight, Fitness Goal, and Workout Intensity. Based on these inputs, the AI generates a personalized 7-day workout plan.

Another important idea identified was to make the system interactive. After receiving the initial plan, the user should be able to provide feedback. For example, the user may indicate that a particular exercise is difficult or that they want a different workout intensity. FitBuddy can use this feedback to generate an updated workout plan.

The brainstorming stage therefore resulted in the following major objectives:

- Develop an AI-based fitness planning application.
- Generate personalized 7-day workout plans.
- Provide nutrition and recovery recommendations.
- Accept user feedback.
- Generate an updated workout plan based on feedback.
- Store user information and generated plans.
- Provide a simple web interface for users.

Thus, the brainstorming phase established the basic concept, purpose, target functionality, and expected outcome of the FitBuddy application.